

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Name perfectION™ Nitrate ISA

Product No 51344763

Pure substance/mixture Mixture

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

Supplier

Fisher Scientific Korea	Mettler-Toledo GmbH
D5,D6, Incheon Airport Logistics	ANALYTICAL
Complex	Im Langacher 44
150, Gonghangdong-Ro Mapo-Gu 296	CH-8606 Greifensee
Beon-Gil	Switzerland
Jung-Gu, Incheon 400-340	Tel: +41-22-567-53-22
Tel: +82-2-6196-5500	Fax: +41-22-567-53-23
Fax: +82-2-6196-5501	Email: ph.lab.support@mt.com

E-mail address

Chem.KR@thermofisher.com See Above

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
 CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

Made in USA

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Based on available data, the classification criteria are not met

Environmental hazards

Chronic aquatic toxicity

Category 3

SAFETY DATA SHEET

perfectION™ Nitrate ISA

Revision Date 05-Sep-2019

Label Elements

Hazard Statements

H412 - Harmful to aquatic life with long lasting effects

Precautionary Statements

Prevention

P273 - Avoid release to the environment

Disposal

P501 - Dispose of contents/ container to an approved incineration plant

Other Hazards

NFPA

Health
0

Flammability
0

Instability
0

Physical hazards
N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS-No	Index No.	Weight %
Water	No information available	7732-18-5	KE-35400	70 - 80
Ammonium Sulfate	No information available	7783-20-2	KE-01743	20 - 30

SECTION 4: FIRST AID MEASURES

First aid measures

General Advice

If symptoms persist, call a physician.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Inhalation

Move to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

Ingestion

Clean mouth with water and drink afterwards plenty of water.

Self-Protection of the First Aider

No special precautions required.

Most important symptoms and effects, both acute and delayed

Most important symptoms and effects

No information available

Indication of any immediate medical attention and special treatment needed

SAFETY DATA SHEET

perfectION™ Nitrate ISA

Revision Date 05-Sep-2019

Notes to Physician Treat symptomatically

SECTION 5: FIREFIGHTING MEASURES

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

Specific Hazards Arising from the Chemical

No information available.

Explosion Data

Sensitivity to Mechanical Impact None

Sensitivity to Static Discharge None

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Personal Precautions

Ensure adequate ventilation. Use personal protective equipment.

Environmental Precautions

Do not flush into surface water or sanitary sewer system.

Methods and Material for Containment and Cleaning Up

Methods for Containment

Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up

Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Handling

Wear personal protective equipment
Ensure adequate ventilation
Do not get in eyes, on skin, or on clothing
Avoid ingestion and inhalation

Conditions for Safe Storage, Including any Incompatibilities

Storage

Keep container tightly closed in a dry and well-ventilated place
Store at room temperature in the original container
Keep away from direct sunlight

Incompatible Products

No information available

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS-No	Korea	ACGIH TLV	OSHA PEL
Water	7732-18-5	Not listed	Not listed	Not listed
Ammonium Sulfate	7783-20-2	STEL: 20 mg/m ³ TWA: 10 mg/m ³	Not listed	Not listed

SAFETY DATA SHEET

perfectION™ Nitrate ISA

Revision Date 05-Sep-2019

Component	CAS-No	European Union	The United Kingdom	Germany
Water	7732-18-5	Not listed	Not listed	Not listed
Ammonium Sulfate	7783-20-2	Not listed	Not listed	Not listed

Biological Exposure Indices

Component	CAS-No	Biological Exposure Indices
Water	7732-18-5	Not listed
Ammonium Sulfate	7783-20-2	Not listed

Appropriate engineering controls

Engineering Measures None under normal use conditions

Individual protection measures, such as personal protective equipment

Personal protective equipment Use only those certified by the Korea Occupational Safety and Health Administration.

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:. Face-shield.

Skin and Body Protection Wear protective gloves/clothing.

Respiratory Protection None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls Prevent product from entering drains Do not allow material to contaminate ground water system

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical State	Liquid
Appearance	Clear
Odor	None
Odor Threshold	No information available
pH	6.4
PH Range	4.3 - 8.5

Property	Values	Remarks • Method
Melting point/freezing point	No information available	
Boiling Point/Range	~ 100 °C / 212 °F	
Flash Point (High in °C)	N/A	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble in water	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature	-	
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	

SAFETY DATA SHEET

perfectION™ Nitrate ISA

Revision Date 05-Sep-2019

Dynamic viscosity	No information available
Explosive Properties	No information available
Oxidizing Properties	No information available

Other Information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

Reactivity

No Information available

Chemical Stability

Stable under normal conditions

Possibility of Hazardous Reactions

None under normal processing

Conditions to Avoid

Extremes of temperature and direct sunlight.

Incompatible Materials

No information available

Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

Product Information

Information on likely routes of exposure

Inhalation	Not an expected route of exposure.
Ingestion	No known effect based on information supplied.
Eyes	Not an expected route of exposure.
Skin	No known effect based on information supplied.

Information on Toxicological Effects

Symptoms	No information available
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Delayed and immediate effects as well as chronic effects from short and long-term exposure

Sensitization	No information available
Mutagenic Effects	No information available
Carcinogenicity	No information available.
Reproductive Effects	No information available
STOT - single exposure	No information available

SAFETY DATA SHEET

perfectION™ Nitrate ISA

Revision Date 05-Sep-2019

STOT - repeated exposure No information available
Target Organs None known.

Aspiration hazard No information available

Numerical measures of toxicity - Product Information

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral) 10923 mg/kg

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity

0% of the mixture consists of components(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Ammonium Sulfate 7783-20-2	-	LC50: = 18 mg/L, 96h (Cyprinus carpio) LC50: 32.2 - 41.9 mg/L, 96h flow-through (Oncorhynchus mykiss) LC50: 5.2 - 8.2 mg/L, 96h static (Oncorhynchus mykiss) LC50: > 100 mg/L, 96h (Pimephales promelas) LC50: 123 - 128 mg/L, 96h semi-static (Poecilia reticulata) LC50: = 126 mg/L, 96h (Poecilia reticulata) LC50: 460 - 1000 mg/L, 96h static (Leuciscus idus) LC50: = 420 mg/L, 96h semi-static (Brachydanio rerio) LC50: = 250 mg/L, 96h (Brachydanio rerio) LC50: = 480 mg/L, 96h flow-through (Brachydanio rerio)	EC50: = 423 mg/L, 24h (Daphnia magna) LC50: = 14 mg/L, 48h (Daphnia magna)

Persistence and Degradability

No information available

Bioaccumulation/ Accumulation

No information available

Mobility

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Component	log Pow
Ammonium Sulfate 7783-20-2	-5.1

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste Disposal Methods Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

SAFETY DATA SHEET

perfectION™ Nitrate ISA

Revision Date 05-Sep-2019

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport	Not Regulated
IATA	Not regulated
IMDG/IMO	Not regulated
Marine Pollutant	Not Applicable
Special Precautions for User	No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS-No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	AICS
Water	7732-18-5	KE-35400	X	231-791-2	X	X	-	X	-	X
Ammonium Sulfate	7783-20-2	KE-01743	X	231-984-1	X	X	-	X	X	X

Component	CAS-No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS-No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Water	7732-18-5	Listed	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Listed	Not applicable	Not applicable

Korean National Regulations

Component	CAS-No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Water	7732-18-5	Annex 1 - KE-35400 Exempt (Index No. 25)	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Annex 1 - KE-01743	Not applicable	Not applicable

Component	CAS-No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	Not applicable	Not applicable

Component	CAS-No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	Not applicable	Not applicable

Component	CAS-No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance

51344763

SAFETY DATA SHEET

perfectION™ Nitrate ISA

Revision Date 05-Sep-2019

Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	STEL: 20 mg/m ³ TWA: 10 mg/m ³	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS-No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

Component	CAS-No	Korea	Biological Exposure Indices
Water	7732-18-5	Not listed	Not listed
Ammonium Sulfate	7783-20-2	STEL: 20 mg/m ³ TWA: 10 mg/m ³	Not listed

US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS-No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Water	7732-18-5	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	Not applicable

CERCLA Not applicable

Component	CAS-No	CERCLA EHS RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Sulfate	7783-20-2	Not applicable	Not applicable	1.0

CLP Classification - Regulation (EC) No 1272/2008

Based on available data, the classification criteria are not met.

H412 - Harmful to aquatic life with long lasting effects.

SAFETY DATA SHEET

perfectION™ Nitrate ISA

Revision Date 05-Sep-2019

P273 - Avoid release to the environment. P501 - Dispose of contents/ container to an approved waste disposal plant.

SECTION 16: OTHER INFORMATION

Prepared By	Thermo Fisher Scientific© Water and Lab Products 22 Alpha Road Chelmsford, MA 01824, USA 1-978-232-6000
Prepared For	Mettler-Toledo GmbH Analytical
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Revision Date	05-Sep-2019
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MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

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End of Safety Data Sheet